# Solution Requirements

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| **Date** | 26 june 2026 |
| **Team ID** | xxxxxx |
| **Project Name** | Opticrop- Smart Agricultural Production Optimization Engine |
| **Maximum Marks** | 4 Marks |

**Define Functional and Non-Functional Requirements:**

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

**Step 1: Functional Requirements (FR)**

| S.No | Requirement Category | Requirement Description | Priority (High/Medium/Low) |
|---|---|---|---|
| 1 | Authentication | Users (farmers, researchers, and policymakers) will securely log in using an email/phone number and password to save their query history and access personalized data. | high |
| 2 | Authorization levels | The system will define specific roles: 'Farmer' (access to Scenarios 1 & 2 for crop recommendations), 'Researcher' (access to Scenario | high |

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| | | 3 for analytics), and 'Admin' (system configuration). | |
| **3** | External interfaces | he system will integrate with external Weather APIs to optionally fetch real-time climate data (temperature, humidity, rainfall) if the user does not input it manually. | medium |
| | | *services the system must connect to]* | |
| **4** | Transactions processing | The core system will handle incoming environmental parameters (N, P, K, pH, etc.), process them through the Scikit-learn ML models, and output a structured recommendation response | high |
| **5** | Reporting | The system will generate printable yield prediction reports for farmers and interactive data-driven dashboards displaying crop-environment relationships for policymakers | medium |
| **6** | Business rules, etc. | *Business rules, processThe recommendation engine must strictly validate that the inputted soil and environmental data fall within scientifically viable agricultural thresholds before predicting a crop.* | high |

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| **7** | Compliance to laws or regulations | Compliance to laws or regulationsThe application must comply with regional data privacy laws regarding the collection and storage of a farmer's geolocation and personal agricultural data. | medium |
| **8** | *Other* | Data Validation)The system must feature strict front-end validation on the Flask web interface to ensure no alphabetical characters are submitted in numeric input fields (like pH or Nitrogen levels) | medium |

**Step 2: Non-Functional Requirements (NFR)**

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| **1** | Performance & Speed | The system must process the ML inference and return crop recommendations quickly to ensure a smooth user experience | ML inference and page load in < 3 seconds |
| **2** | Scalability | The Flask backend must be able to handle growth in users, especially during peak farming and planting seasons | Supports up to 5,000 concurrent users without performance degradation. |
| **3** | Security & Data Privacy | *All user passwords must be hashed, and the agricultural datasets and personal information must be protected during transit.]* | HTTPS protocol used for all web traffic; AES-256 encryption for database records. |

| **S.No** | **NFR Category** | **Requirement Description** | **Target Metric / Acceptance Criteria** |
|---|---|---|---|

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|---|---|---|---|
| **4** | Reliability & Availability | *The platform must maintain high uptime so farmers and researchers can access vital agricultural insights whenever needed.]* | *99.9% uptime per month* |
| **5** | Usability & Accessibility | The user interface must be simple enough for users with limited technical expertise and must function on mobile devices used in the field. | *Mobile esponsive UI design; complies with WCAG 2.1 AA accessibility standards.)* |
| **6** | *Other* | *The software must be easily deployable and operable across different operating systems as specified in the project requirements.]* | Application runs seamlessly on Windows, Linux, and macOS environments. |